

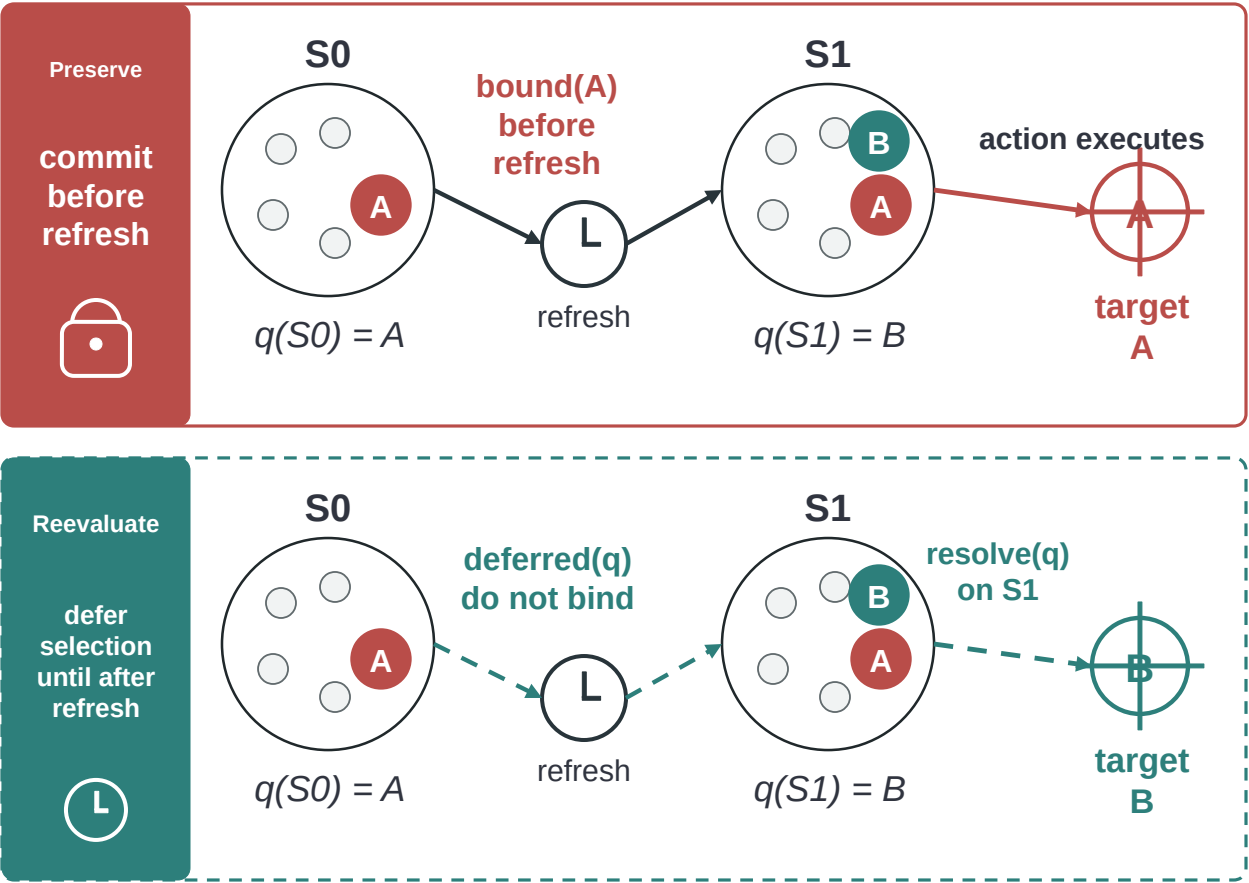
A. Construct the matched pair

Fixed: S0, S1, q, action, schema, controller interface

Unique change: instruction commitment timing

Shared transition: $q(S0) = A \rightarrow \text{refresh} \rightarrow q(S1) = B$

A remains present and action-valid in S1



What is fixed across the pair

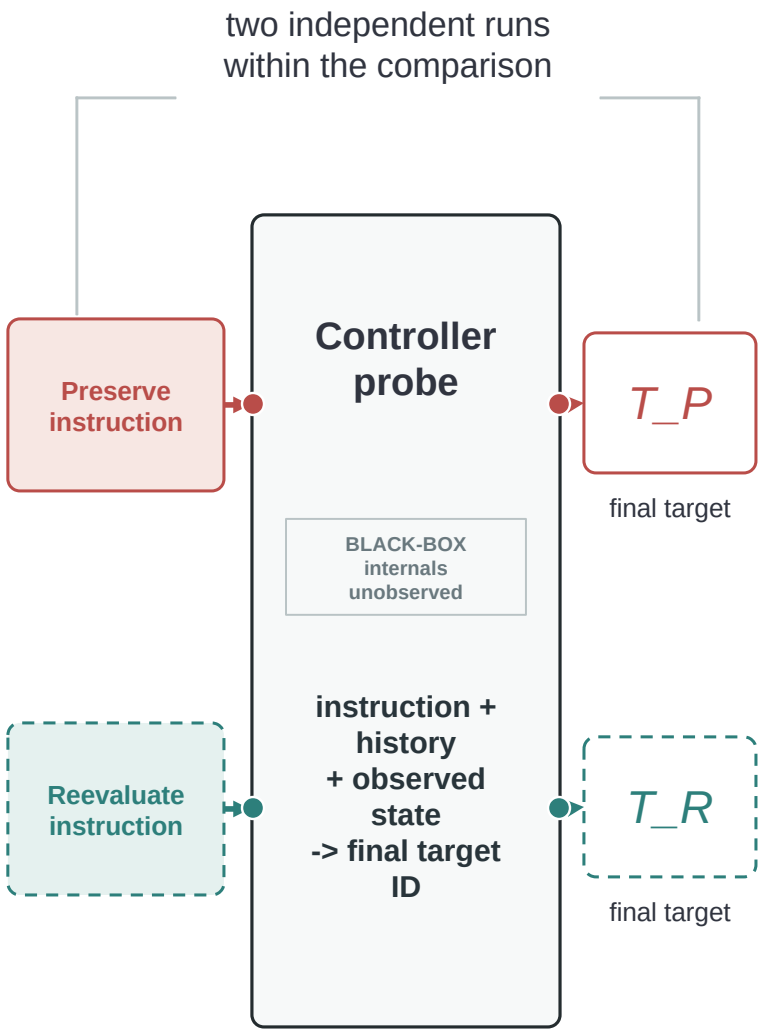
- ✓ States: S0, S1, and all entities
- ✓ Selector: q
- ✓ Action and schema
- ✓ Controller interface

What is uniquely changed

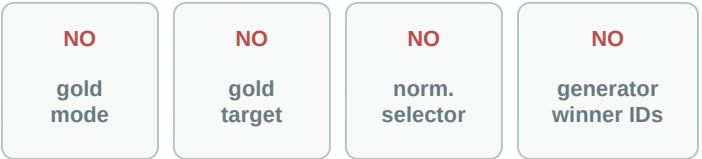
Instruction commitment timing
(bind before refresh vs. defer until after refresh)

B. Run the pair

same controller + same interface



Withheld from the probe



Observed instruction, history, and state follow the tested controller interface.

C. Observable readouts

1 Pair accuracy (PairAcc)

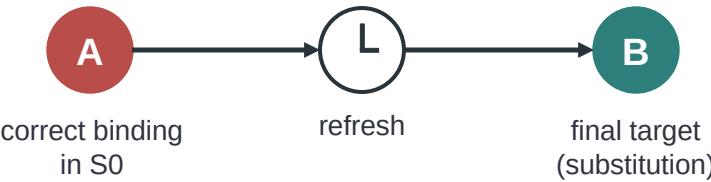
Unit = one complete changed-winner pair



Denominator: all complete changed-winner pairs

2 Conditional substitution

Focus = replacement after a correct observable binding



Denom. (Preserve): correct binding + completed refresh + changed winner + surviving, action-valid A

3 Execution subset (tool writes)

Trace = selected target ID -> tool write -> state diff



Denominator: executed model-issued writes (subset of all evaluation rows)